

1 Solution — GIAM 1.2

Problem 2

1.1 Problem

Use the sieve of Eratosthenes to find all prime numbers up to 100.

1.2 How Far Do We Have to Sieve?

The textbook flags the key bound in Section 1.2:

The sieve gives us a means of finding all the primes up to p^2 by using the primes up to (but not including) p .

Since $100 = 10^2$, we only need to sieve by primes strictly less than 10 — that is, by

2, 3, 5, 7.

Any composite $n \leq 100$ has a prime factor $\leq \sqrt{n} \leq 10$, so the smallest such factor is one of 2, 3, 5, 7. After we cross out all multiples of these four primes (other than the primes themselves), every remaining number ≥ 2 is prime.

1.3 The Four Passes

Pass	Smallest prime p	Numbers crossed out (multiples of p not yet crossed)	Count
1	2	4, 6, 8, 10, 12, $\dots$, 100 — all even integers > 2	49
2	3	9, 15, 21, 27, 33, 39, 45, 51, 57, 63, 69, 75, 81, 87, 93, 99 — odd multiples of 3	16
3	5	25, 35, 55, 65, 85, 95 — odd multiples of 5 not yet crossed	6

4	7	49, 77, 91	— multiples of 7	coprime to 2, 3, 5	3
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Table 1.1

After pass 4 we have crossed out a total of $49 + 16 + 6 + 3 = 74$ composites. Together with 1 (which is neither prime nor composite), that accounts for 75 of the 100 integers. The remaining 25 are exactly the primes ≤ 100 .

1.4 The Sieve in One Picture

Sieve of Eratosthenes: primes ≤ 100 (green) with composites coloured by smallest prime factor

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

 prime (25)	 $\div 2$ (49)	 $\div 3$ (16)	 $\div 5$ (6)	 $\div 7$ (3)
 1 — neither prime nor composite				

Sieve by primes $\leq \sqrt{100} = 10$ (i.e. 2, 3, 5, 7); $49 + 16 + 6 + 3 = 74$ composites; 25 primes; 1 special

Figure 1.1: Primes (green, 25 of them) and composites coloured by the smallest prime that crosses them out: yellow = $\div 2$ (49), orange = $\div 3$ (16),

pink = $\div 5$ (6), purple = $\div 7$ (3). The number 1 (grey) is neither prime nor composite.

1.5 The Primes — All 25 of Them

Reading the green cells from the figure (or equivalently, listing what survives the four passes):

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

There are **25** primes less than or equal to 100.

1.6 Remark — The Prime-Counting Function $\pi(n)$

The count "25 primes up to 100" is a value of the **prime-counting function** $\pi(n)$, defined as the number of primes $\leq n$:

$$\pi(100) = 25.$$

The function grows surprisingly regularly. A few values:

n	10	100	1000	10^4	10^6	10^9
$\pi(n)$	4	25	168	1229	78498	50847534

Table 1.2

The **Prime Number Theorem** (Hadamard and de la Vallée Poussin, 1896) makes this regularity precise:

$$\pi(n) \sim \frac{n}{\ln n} \quad \text{as } n \rightarrow \infty.$$

For $n = 100$: $n/\ln n \approx 100/4.605 \approx 21.7$ — the theorem under-estimates the actual count by about 13%. By $n = 10^9$ the estimate is within 5% of the true value. The PNT's proof is far beyond a Chapter-1 exercise — it uses complex analysis and the Riemann zeta function — but the *statement* is easy to appreciate from the sieve picture.

1.7 Remark — Twin Primes Up to 100

A **twin prime** pair is two primes that differ by 2. Reading down the list:

$$(3, 5), (5, 7), (11, 13), (17, 19), (29, 31), (41, 43), (59, 61), (71, 73)$$

— eight twin-prime pairs up to 100. (Some authors exclude (3, 5) since 3 and 5 are not both odd-greater-than-3 primes, but the count remains in the same ballpark either way.)

Twin primes are famously *not* known to be infinite — the **twin prime conjecture** says there are infinitely many such pairs, and remains unproved as of writing. Yitang Zhang’s 2013 breakthrough showed primes “infinitely close” in a quantitative sense (gaps of at most 7×10^7), and follow-up work has dropped the bound to 246. But “ p and $p + 2$ both prime infinitely often” is still open.

1.8 Remark — Prime Gaps Within 1–100

The **prime gap** g_n is the difference between consecutive primes. From the list:

$$g_1 = 1 \ (2 \rightarrow 3), \ g_2 = 2 \ (3 \rightarrow 5), \ g_3 = 2 \ (5 \rightarrow 7), \ g_4 = 4 \ (7 \rightarrow 11), \dots$$

The largest gap below 100 is 8, between the primes 89 and 97 — a famously prominent stretch on the chart. No primes lie in $\{90, 91, 92, 93, 94, 95, 96\}$: 90 is even, 91 = $7 \cdot 13$, 92 even, 93 = $3 \cdot 31$, 94 even, 95 = $5 \cdot 19$, 96 even. Seven consecutive composites — the longest run in 1–100.

1.9 Remark — Why the Sieve Stops at 7

We don’t need to sieve by 11 because $11^2 = 121 > 100$. Any composite ≤ 100 has *some* prime factor $\leq \sqrt{100} = 10$, so after crossing out multiples of 2, 3, 5, 7, every remaining number is either 1 or itself prime.

A complementary intuition: if $n \leq 100$ has no prime factor ≤ 10 , then n has *at least two* prime factors > 10 , and the product of any two such primes is > 100 . So n cannot have two such factors — i.e. n has at most one prime factor > 10 , and if it has any then n is that prime.

So sieving by **2, 3, 5, 7** is *sufficient* — and any further sieving (by **11, 13, ...**) would do nothing new because there are no remaining un-crossed composites.

1.10 Remark — Why 1 is Not Prime

The grey cell at $n = 1$ is a notational convenience. The textbook’s definition reads “a prime is a positive integer **greater than 1**” — the explicit exclusion of **1** is necessary for the **Fundamental Theorem of Arithmetic** to hold: if **1** counted as prime, every prime factorisation could be padded with extra **1**s ($6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3 = \dots$), breaking uniqueness.

Equivalently: the alternative “Euclid’s lemma” definition (“ p divides ab implies $p \mid a$ or $p \mid b$ ”) *would* be satisfied by **1** — but only trivially. We exclude **1** because the resulting theorems are cleaner, not because **1** “fails” any algebraic criterion.

1.11 Remark — Visual Cues in the Figure

A few patterns leap out from the colored grid:

- **Even columns are nearly all yellow** (smallest factor **2**). Only **2** itself in column 2 escapes.
- **Column 5 (and 0, i.e. the column of multiples-of-10) is yellow or pink**: column 0 (numbers ending in 0) is all yellow (even); column 5 starts with **5** (green) and then alternates pink (**25, 35, ...**) and yellow (**50, 60, ...**).
- **All primes ≥ 5 end in 1, 3, 7, or 9**: columns 1, 3, 7, 9. Numbers ending in 0, 2, 4, 6, 8 are even (yellow except **2**), and numbers ending in 5 are multiples of 5 (pink except **5**).
- **The purple cells (49, 77, 91) are scattered**. These are the only composites ≤ 100 whose smallest prime factor is **7** — a very thin set.

These visual regularities reflect the *modular arithmetic* underlying primality: the “shape” of primes is constrained by their residues mod **2, 3, 5**.

1.12 Remark — Beyond 100: $\pi(n)$ Grows Steadily

If we extended the sieve to higher n , the rate at which new primes appear *slows*. Below are the gaps between successive values of $\pi(10^k)$:

n range	new primes in this decade	density (per integer)
1–10	4	40%
11–100	21	$\approx 23\%$
101–1000	143	$\approx 16\%$
1001–10 000	1061	$\approx 12\%$
10^4 – 10^5	8363	$\approx 9\%$

Table 1.3

Primes thin out, but never disappear (Euclid’s theorem: infinitely many primes). The 1-in-25 rate visible in our grid is already lower than the 1-in-2.5 rate in the first decade, and will continue dropping like $1/\ln n$.